Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/28/2022
PI Name:	Limoli, Charles Ph.D.		
Project Title:	NSCOR: Mechanisms Underlying Charged Particle-Induced Disruption of CNS Function		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-NSCOR Radiation
Start Date:	07/01/2015	**End Date:**	06/30/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2023 per NSSC information (Ed., 2/14/23) NOTE: End date changed to 2/28/2023 per NSSC information (Ed., 8/6/22) NOTE: End date changed to 2/28/2022 per NSSC information (Ed., 3/12/21) NOTE: End date changed to 2/28/2021 per NSSC information (Ed., 4/6/2020)		
Key Personnel Changes/Previous PI:	Per the PI, Dr. Parihar is no longer on this project and is currently working in India (Ed., 8/7/22).		
COI Name (Institution):	Baulch, Janet Ph.D. (University of California, Irvine) Britten, Richard Ph.D. (Eastern Virginia Medical School) Soltesz, Ivan Ph.D. (University of California, Irvine) Ascharya, Munjal Ph.D. (University of California, Irvine)		
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Task Description:	Humans exposed to the space radiation environment encounter a different type of radiation than that found on Earth. Highly energetic particles can traverse the body, including the brain, and leave a path of damage that is difficult to repair and disrupts function. One consequence of this exposure is the development of radiation-induced cognitive dysfunction. The focus of this application is to resolve the mechanisms leading to radiation-induced cognitive dysfunction, in order to improve risk estimates for early and late CNS (central nervous system) effects caused by exposure to the space radiation environment. We have found that radiation impacts excitatory and inhibitory circuitry in the brain differently, and leads to significant disruptions to the structural integrity of neurons while altering the levels and activity of the proteins that regulate neuronal function. The situation is further complicated by the fact that different regions of the brain exhibit differential sensitivity to charged particle irradiation. Based on the foregoing we propose a comprehensive series of mechanistic studies analyzing the biochemical, structural, and electrophysiologic changes induced by radiation exposure that alter function to impact cognition. Carefully selected cognitive tasks and a wide variety of approaches will be used to critically analyze microcircuits in the brain and how charged particle-induced changes are transmitted via unique signaling pathways between cells that can alter gene expression profiles within cells. Collectively these studies will span a radiation effects pathway from synapse to cognition, with an overarching goal of determining whether charged particle irradiation leads to alterations in the excitatory and inhibitory activity of the brain at a global and/or region specific level. Our studies will therefore elucidate the functional consequences of radiation-induced changes on cognition that impose risk on mission critical activities and long-term cognitive health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Work will characterize several of the adverse effects of ionizing radiation on the CNS and help to define the early and late CNS risks associated with exposure to the space radiation environment. There is a real possibility that many of the consequences of cosmic radiation exposure will extend over many years and impact (possibly negatively) neurocognitive health after return to Earth. Thus, research elucidating the mechanistic basis of charged particle-induced effects in the irradiated brain may well have functional importance to terrestrial situations where the public may be exposed to ionizing radiation from a variety of potential sources.
Task Progress:	Complex 33-beam GCR simulation: Our prior report detailed our significant effort in moving this large concerted and multi-institutional study forward last year. We are now on the verge of publishing much of these findings but still require additional time to finalize many of the data sets we detailed in our previous report. The following is a succinct re-cap of where things currently stand: a) Manuscript under secondary review at Cellular and Molecular Sciences: Alaghband, Y., Klein, P.M., Shelerud, L.M., Kane, A.E. Ngoc-Lien Doan, N-L., Ru, N., Acharya, M.M., Kramár, E.A., Wood, M.A., Sinclair, D.A., Soltesz, I., Limoli, C.L and Baulch, J.E. Multifaceted neurocognitive impairments derived from complex 33-beam simulated galactic cosmic radiation exposure. Submitted, Cell Mol Sci (2022). Using the complex 33-beam galactic cosmic radiation simulations (GCR Sim), we provide new insights into the response of male and female mice exposed to acute (1 day) and chronic (1 month) exposures (40-50 cGy) relevant to deep space travel. In particular, the results of the Object in Updated Location task suggested that either acute or chronic exposure to GCR Sim induced persistent impairments in hippocampus-dependent memory formation and reconsolidation in female mice that did not manifest robustly in irradiated male mice. Interestingly, irradiated male mice were impaired in the recognition of object novelty that was not observed in females. Further, male mice exposed to chronic GCR Sim exhibited increased aggressive behavior on the tube dominance test. Given the observed hippocampus-related neurobehavioral impairments, electrophysiology studies were used to evaluate synaptic plasticity in the hippocampal CA1 region. Significant reductions in long-term potentiation (LTP) were found after each irradiation paradigm (chronic > acute) and in both sexes, compared to unirradiated controls. Interestingly, network level disruptions did not translate to altered intrinsic electrophysiological properties of CA1 pyramidal cells, whereas acute exposures caused modest drops in excitatory synaptic signaling in male mice. Ultrastructural analysis of synapses in the hippocampal CA1 region found smaller post synaptic densities (PSDs) in the larger spines of chronic exposed mice compared to controls and acute exposed mice. Myelination was also affected by GCR irradiation with acute GCR exposed mice having less myelination on small caliber (<0.3 mm) and larger (>0.5 mm) axons. Present findings provide further evidence that space relevant radiation exposures disrupt critical cognitive processes and underlying neuronal network level plasticity, albeit not to the extent that might have been predicted from earlier work. b) A subset of animals subjected to the complex 33-beam galactic cosmic radiation simulations (GCR Sim) has also been analyzed in collaboration with Raj Desai and Brian Kangas at Harvard University. A manuscript is currently under preparation with brief highlights provided below: i) An Economic Demand behavioral test revealed no effects of GCR on reward sensitivity. ii) A Psychomotor Vigilance Task showed significant deleterious effects of both acute and chronic GCR. Adverse effects subside over time (i.e., extended testing conditions), but performance never fully approximated that of unirradiated control mice. These cohorts were subsequently analyzed for neurotransmitter analysis by in-dwelling microdialysis chambers; data were as follows: iii) Chronic GCR Exposure increased Glutamate, GABA, dopamine, serotonin, and norepinephrine sensitivity (releasable pool) vs controls. iv) Acute GCR Exposure increased dopamine responsiveness vs. controls. v) Chronic GCR exposure increases sensitivity in multiple neurotransmitter systems in the prefrontal cortex (PFC), whereas acute GCR appears to primarily impact dopamine in the PFC – a brain region that plays a key role in complex behavior and cognition. vi) Lastly, a complete RNAseq analysis was undertaken in conjunction with these studies at Stanford University. Principal component analysis separated at three cohorts (control, acute GCR, chronic GCR). Once the bioinformatic and pathway analysis is complete, this manuscript will be finalized for submission. c) In a final study nearing completion at Indiana University (in collaboration with Dr. John Beggs), a study of criticality using a 256-electrode array to analyze synaptic hubs and nodes that fire in oscillatory synchrony has been undertaken. Preliminary results have found surprisingly that net neurotransmission is increased in the irradiated cohorts, but in a pattern that indicates synaptic dysregulation, such that a scrambling of normal synaptic connections has occurred in response to GCR irradiation. This data will be finalized for submission over the requested No-Cost Extension period (NCE). Pion exposures: As detailed in our prior report, we have been engaged in some pion exposures, to assess the possibility that such exposures pose a threat to the functionality of the central nervous system (CNS). Our behavioral studies

	exposing mice to 0, 5, and 20 cGy of positive and negative pions has been completed. We have presented these findings at several recent meetings and found that all associative recognition patterns of learning and memory on the Objects in Updated Location (OUL) task were found to be significantly impaired. Additional tasks of Novel Object Recognition (NOR) and Fear Extinction (FE) also returned significant impairments in the irradiated cohorts, with the most significant impact found with the negative versus positive pions exposures. Importantly, these data were obtained over a 1-year post-irradiation interval, and point to the importance of further investigations where the dose equivalent of pions for eliciting functional change in the irradiated CNS is far greater than expected for typical low let photon modality. As a follow up to the behavioral assessment, we have also undertaken a complete RNAseq analysis, and that data is currently being processed to determine if any pion-specific signatures can be identified in the irradiated brains isolated after 1-year. Finalizing this important data set will be a major goal planned during the NCE.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Klein PM, Alaghband Y, Doan NL, Ru N, Drayson OGG, Baulch JE, Kramár EA, Wood MA, Soltesz I, Limoli CL. "Acute, low-dose neutron exposures adversely impact central nervous system function." Int J Mol Sci. 2021 Aug 21;22(16):9020. https://doi.org/10.3390/ijms22169020 ; PMID: 34445726; PMCID: PMC8396607 , Aug-2021
Articles in Peer-reviewed Journals	Britten RA, Fesshaye A, Ihle P, Wheeler A, Baulch JE, Limoli CL, Stark CE. "Dissecting differential complex behavioral responses to simulated space radiation exposures." Radiat Res. 2021 Dec 17;197(3):289-297. https://doi.org/10.1667/RADE-21-00068.1 ; PMID: 34905619 , Dec-2021
Articles in Peer-reviewed Journals	Schaeffer EA, Blackwell AA, Osterlund Oltmanns JA, Einhaus R, Lake R, Hein CP, Baulch JE, Limoli CL, Ton ST, Kartje GL, Wallace DG. "Differential organization of open field behavior in mice following acute or chronic simulated GCR exposure." Behav. Brain Res. 2022 Jan;416:113577. https://doi.org/10.1016/j.bbr.2021.113577 , Jan-2022